

Some Facts

1. India, on annual basis, has only managed to collect ten percent of the total e-waste it generated
2. India is the third most e-waste generating country with 3.2 million tonnes in 2019.
3. Global E-waste Monitor 2020 estimates that in 2019 only 17.4 percent of e-waste was collected and recycled leading to a loss of nearly \$47 billion.
4. Gold, silver, copper, platinum are some examples of high-value materials that can be recovered from e-waste.
5. Establishing a good e-waste collection channel or system is the biggest problem as well as an opportunity.
6. Refurbishment and recycling are two industries that can complement each other well.
7. There is enough room for startups as well as big brands to venture into this field as it is highly unorganised at present.
8. Recycling offsets the impact of mining on nature by enabling a circular economy in electronics.

in electronics,” says Pranshu Singhal, founder of Karo Sambhav. He is an electronic waste PRO offering solutions for e-waste management and EPR services in India.

Niti Ayog, the public policy think tank of the government of India, has made the first move and entrusted the Ministry of Electronics and IT (MeitY) to formulate a plan that would encourage India to work better around the e-waste it generates. The move comes on the cusp of the fact that India is the third largest consumer of raw materials produced globally and would be consuming nearly 15 billion tonnes of material by 2030 with the current economic trends.

“We have really made a good start. Not a lot of countries have legislations identifying the problem of e-waste and trying to solve that problem. It is good to see the current government understanding that there is a problem, and it needs to be addressed. In terms of e-waste, there is a long way to go, but with the right policy and implementation, India has the ability to become a superpower in terms of recycling and enabling a superior circular economy around e-waste,” says Karan Thakkar, founder of EcoCentric—a recycling company headquartered in Mumbai.

In terms of elements used in manufacturing electronics, the rate of extraction of abiotic resources including iron, copper, silver, gold, aluminium, manganese, chromium, and zinc along with various other rare earth elements, is significantly higher than the rate of their formation in nature. The simplest of supply and demand laws state that anything that has more demand than what can be supplied tends to get on

the expensive side. This might simply spell that businesses and end consumers will have to pay more in order to source electronics and components as the raw materials used to fabricate them start to become scarce.

However, the key to reversing this trend might be hiding in plain sight under two terms, namely circular economy and recycling. We should not forget that business opportunities around recycling and circular economy are mammoth, and these also possess the capability to help nature replenish elements and stay healthy.

The business opportunities

Before we take a look at the possible e-waste business opportunities knocking on the door, it is important to understand how much and what kinds of waste the country, as well as the world generates. As per Global E-Waste Monitor 2020 report, India generated 3.2 million tonnes of e-waste in 2019, ranking third after China who generated 10.1 million tonnes and the United States that generated 6.9 million tonnes.

The rest of the world, as per the same report, generated a striking 53.6 million tonnes of e-waste, an average of 7.3kg per capita in 2019. The growing amount of e-waste is directly

linked to higher consumption rates of electric and electronic equipment (EEE) manufacturing, short product life-cycles, rapid technological advancement, and few repair options. Asia generated the highest quantity of e-waste in 2019 at 24.9 million tonnes, followed by the Americas (13.1 million tonnes) and Europe (12 million tonnes), while Africa and Oceania generated only 2.9 and 0.7 million tonnes, respectively.

However, India collected just 10 percent of the electronic waste (e-waste) estimated to have been generated in the country in 2018-19 and 3.5 percent of that generated in 2017-18, as per a recent report by the Central Pollution Control Board. During 2018, e-waste collection was just 3.1 percent in India.

“Rural India has just started consuming electronics and the size of rural India is much bigger than that of urban India. The amount of e-waste this section of the country will start generating two to three years down the line will be much more,” says Karan.

If current economic trends persist then India’s material requirements, as per Niti Ayog, are projected to be nearly 15 billion tonnes by 2030 and little above 25 billion tonnes by 2050. For India to fulfill its resource needs, it is imperative that we follow circular economy approach rather than the current linear economy principle of take-make-dispose. The significant gap between the waste generated and the waste collected is evidently clear from the reports but what’s even more clear is the business opportunity e-waste offers.

“I see e-waste just like any other industry where there are business opportunities at every juncture. Right from the start of manufacturing a component to supplying the end product to a consumer, there is opportunity in all the parts of the supply chain of

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